



北京理工大学
BEIJING INSTITUTE OF TECHNOLOGY

27 Fall PhD Application Research Portfolio

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Content

1 | **Self Introduction**

2 | **Research Experience**

3 | **Future Research Interest**

➤ Name: Chengye Wang ➤ Nationality: Chinese

➤ Education Background:

Bachelor Degree (2020.9-2024.06)

Jilin University (Project 985), Changchun, China

Major: Vehicle Engineering

Supervisor: Prof. Nan Xu

GPA: 3.73/4.0, Top 5%

Awards: Outstanding Graduate

Thesis: “Vehicle trajectory planning method based on polynomial curve optimization.”

Postgraduate Student (2024.9-2027.06)

Beijing Institute of Technology (Project 985), Beijing, China

Major: Mechanical Engineering

Supervisor: Prof. Yechen Qin

GPA: 86/100

Awards: Outstanding Freshman Scholarship

Research Interests: Vehicle driving stability analysis, Chassis control of Distributed Drive Electric Vehicle (DDEV), Trajectory planning.

➤ Research Skills:

Matlab/Simulink, CarSim, Python, Ubuntu/ROS, C++, Latex, Automotive hardware and software debug, and Microsoft Office.



Research Background 1: Vehicle Driving Stability Analysis

Unclear stability analysis under multidimensional factors

Driving Factors

complex environment



Control Ability

controller

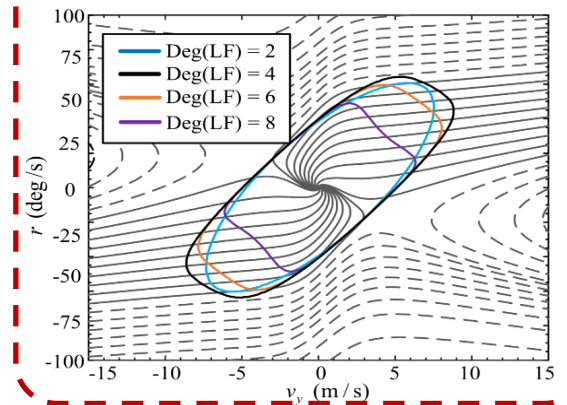


multi-actuator



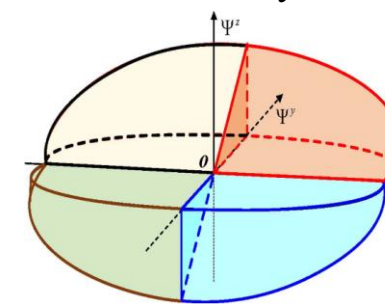
Stability Analysis Method

Lyapunov Function Method



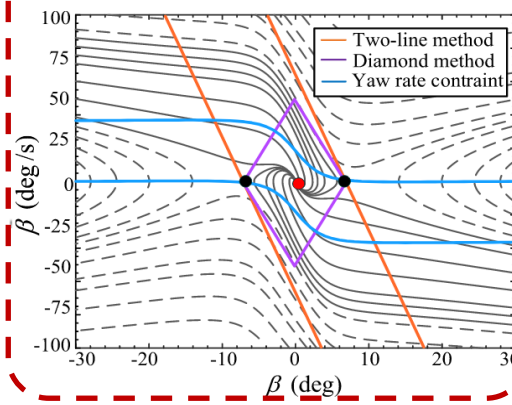
Inability to Model Complex Nonlinearities

Tire Force Analysis Method



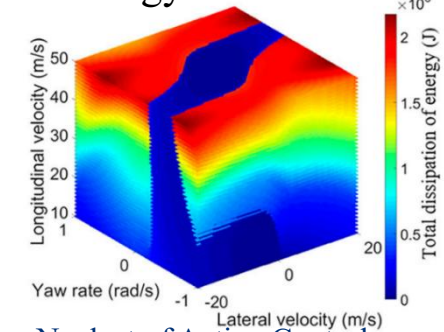
Tire saturation doesn't equal vehicle instability

Phase Plane Method



Based on open-loop assumptions

Energy Method

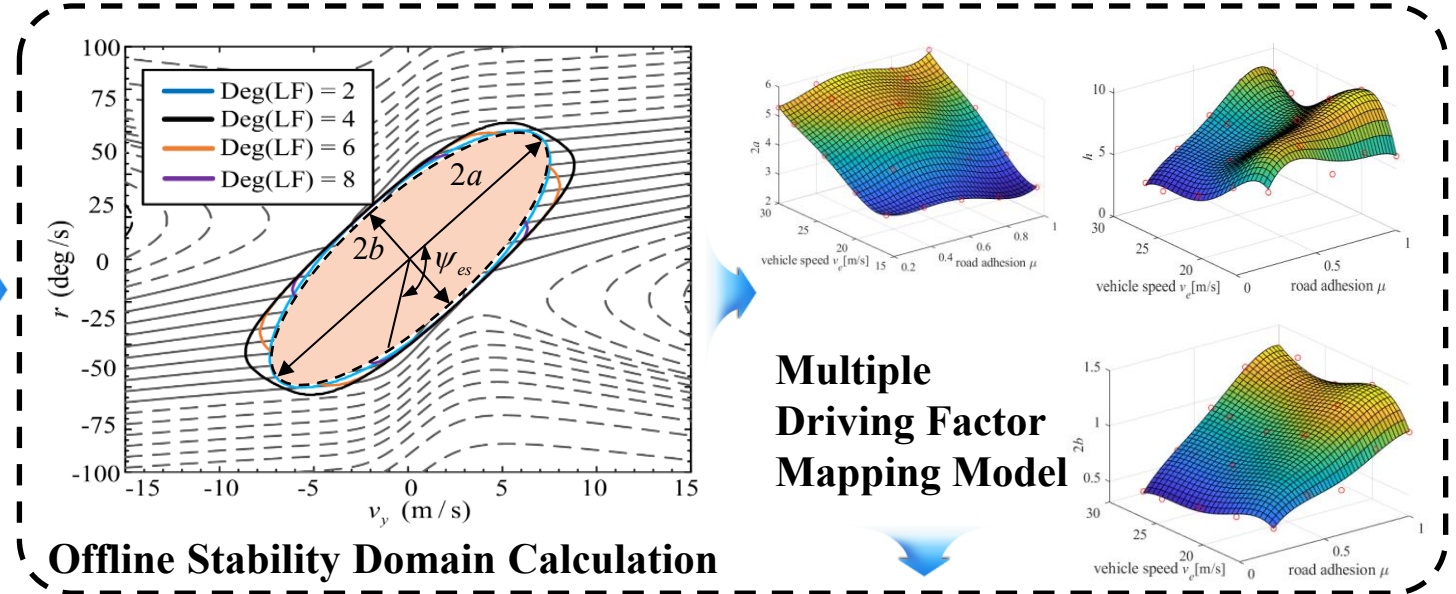
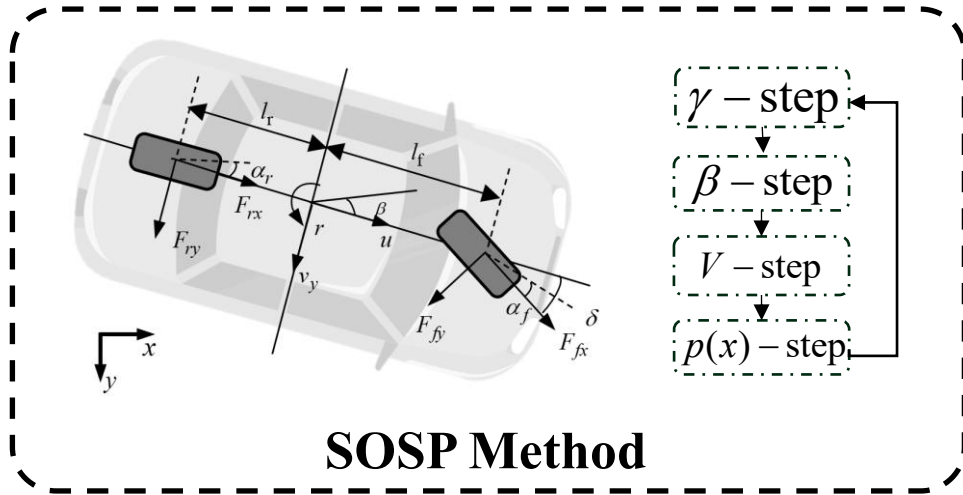


Neglect of Active Control

Physical Parameters, Environment, and Control Ability Jointly Determine the Stability Limit

Research Experience 1: Vehicle Driving Stability Analysis

Lyapunov Function Method-Sum of Squares Programming (SOSP)



$$\text{Elliptical constraints: } g(v_y, r) = \left(\frac{\cos^2 f_{\psi_{es}}}{f_a^2} + \frac{\sin^2 f_{\psi_{es}}}{f_b^2} \right) r + 2 \cos f_{\psi_{es}} \sin f_{\psi_{es}} \times \left(\frac{1}{f_a^2} - \frac{1}{f_b^2} \right) v_y r + \left(\frac{\cos^2 f_{\psi_{es}}}{f_a^2} + \frac{\sin^2 f_{\psi_{es}}}{f_b^2} \right) r - 1$$

Related publications:

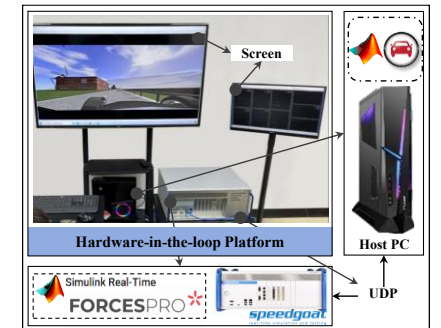
1. Yu Zhang, **Chengye Wang**, Fu Du, Mingming Dong, Yechen Qin, Ming Mao. "Dynamic Region of Stability Integrated Path-tracking Control for Intelligent Vehicles". **Chinese Journal of Mechanical Engineering**. (accepted)
2. **Chengye Wang**, Yu Zhang, Xuepeng Hu, Haipeng Qin, Guoli Wang, Yechen Qin. "Real-time multidimensional vehicle dynamic stability domain calculation and its application in intelligent vehicles", **SAE 2025 Intelligent and Connected Vehicles Symposium**, Chongqing, China. (accepted)

MPC tracking controller

$$u^* = \arg \min J$$

$$\text{s.t. } \dot{x}^{\text{trk}} = f(x, u), g(v_y, r) \big|_{f_a, f_b, f_{\psi_{es}}} \leq 0$$

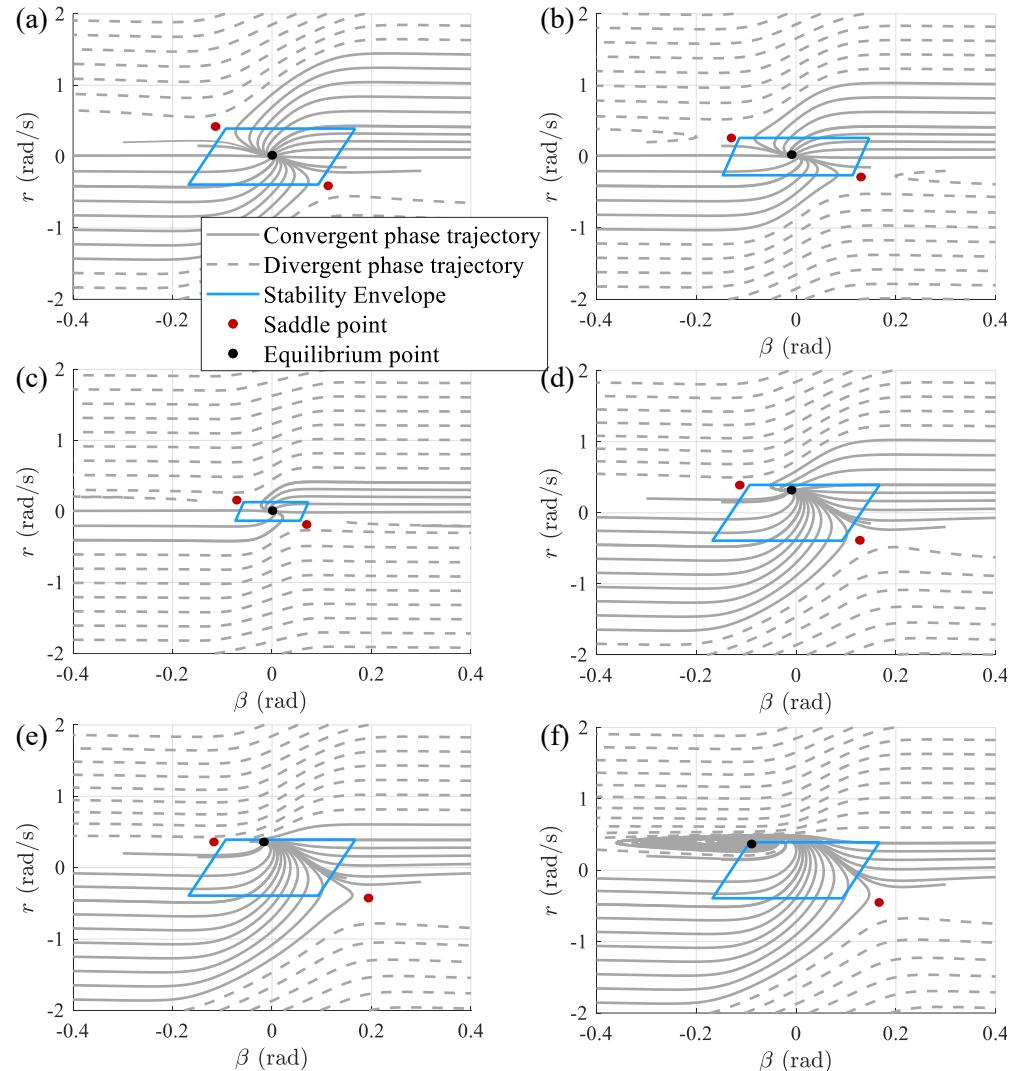
$$u_{\min} \leq u \leq u_{\max}, x_{\min} \leq x \leq x_{\max}$$



Research Experience 1: Vehicle Driving Stability Analysis

■ Analysis of Active Control in the Recovery of the Phase Plane Trajectory

Open-loop Phase Plane

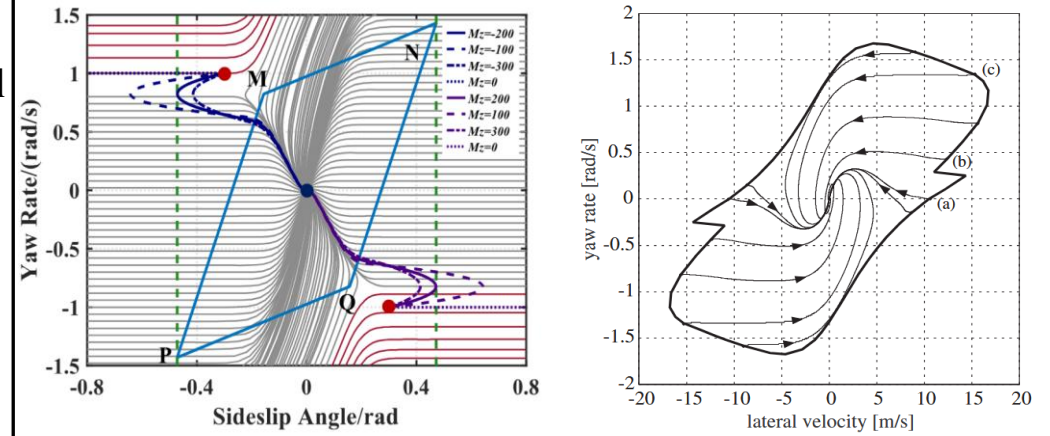


no
control
input

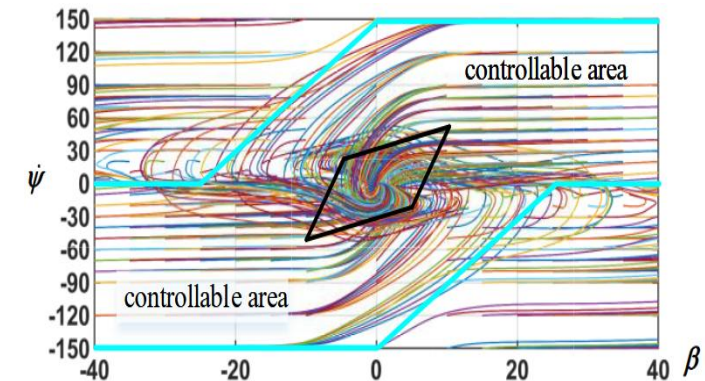
active
control
input

Phase Plane Considering Active Control Inputs

- Phase Trajectory Convergence via Direct Yaw Control (DYC):



- Phase Trajectory Convergence via Active Front Steering (AFS):



Research Experience 1: Vehicle Driving Stability Analysis

■ Specific Control Law – guided Controlled Stability Region (CSR)

Stability Control Law Design

Predictive Model:

$$\begin{cases} \dot{\beta} = \frac{1}{mv_x} [(F_{y,fl} + F_{y,fr}) \cos \delta_f + (F_{y,rl} + F_{y,rr})] - r \\ \dot{r} = \frac{1}{I_z} [l_f (F_{y,fl} + F_{y,fr}) \cos \delta_f - l_r (F_{y,rl} + F_{y,rr}) + \frac{B_f}{2} (F_{y,fl} - F_{y,fr}) \sin \delta_f + M_z^{stab}] \\ \dot{v}_x = \frac{F_{x,all}^{stab}}{m} + v_x \tan \beta \cdot r \end{cases}$$

State/Control/Reference Definition:

$$x^{stab} = [\beta, r, v_x]^\top \quad u^{stab} = [M_z^{stab}, F_{x,all}^{stab}]^\top$$

$$x_{ref}^{stab} = [\beta_{ref} = 0, r_{ref} = \text{sgn}(\delta_f) \min\{v_x \delta_f / (L(1 + Kv_x^2)), \mu g / v_x\}, v_{x,ref} = v_{x0}]^\top$$

Objective Function:

$$J^{stab} = \int_t^{t+T_p} [w_{stab0ref} (x^{stab} - x_{ref}^{stab})^2 + w_{stab0u} (u^{stab})^2 + w_{stab0\Delta u} (\Delta u^{stab})^2] dt$$

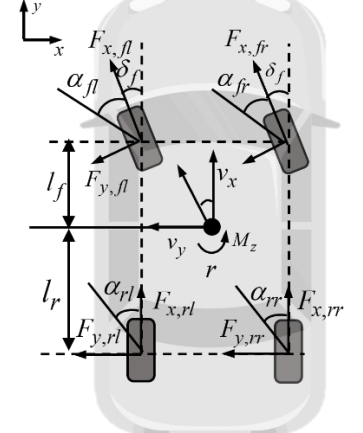
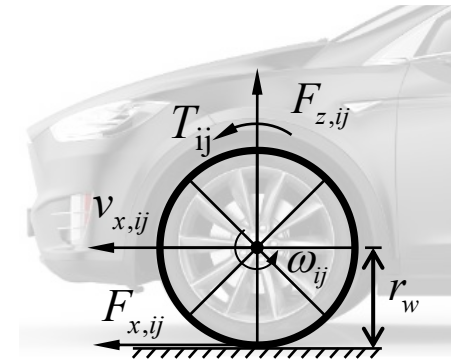
MPC Control Law:

$$u^{stab*} = \arg \min_{u^{stab}} J^{stab}$$

$$\text{s.t. } \dot{x}^{stab} = f(x^{stab}, u^{stab}), u_{min}^{stab} \leq u^{stab} \leq u_{max}^{stab}$$

Controlled Phase Trajectory Calculation

7 DOF Phase Plane Model: $\dot{x} = f(x, u)$



QP

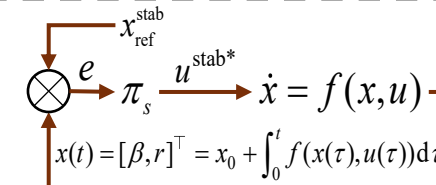
$$x = [\beta, r, v_x, \omega_{ij}]^\top \quad u = [T_{ij}, \delta_f]^\top$$

$$x_0 = [\beta_0, r_0, v_{x0}, \omega_{ij0}]^\top$$

$$x(t) = x_0 + \int_0^t f(x(\tau), u(\tau)) d\tau = \phi(t; x_0, u(\cdot))$$

$$\begin{cases} \dot{v}_y = \frac{1}{m} [F_{y,f} \cos \delta_f + F_{x,f} \sin \delta_f + F_{y,r}] - v_x r \\ \dot{v}_x = \frac{1}{m} [F_{x,f} \cos \delta_f - F_{y,f} \sin \delta_f + F_{x,r}] + v_y r \\ \dot{r} = \frac{1}{I_z} [l_f F_{y,f} \cos \delta_f - l_r F_{y,r} + M_z] \\ \dot{\omega}_{ij} = \frac{T_{ij} - F_{x,ij} r_w}{I_w}, \beta = \arctan\left(\frac{v_y}{v_x}\right) \end{cases}$$

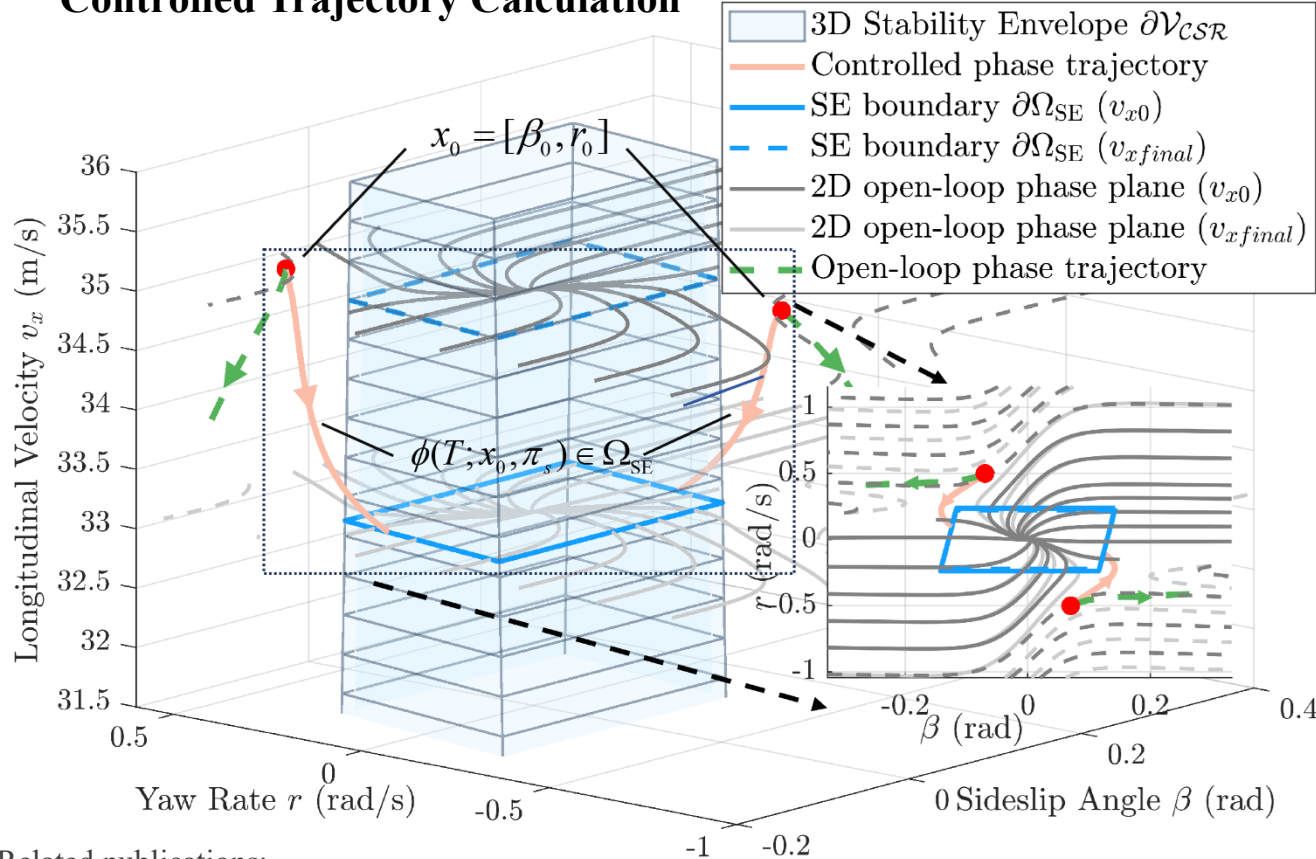
CSR Calculation:



Research Experience 1: Vehicle Driving Stability Analysis

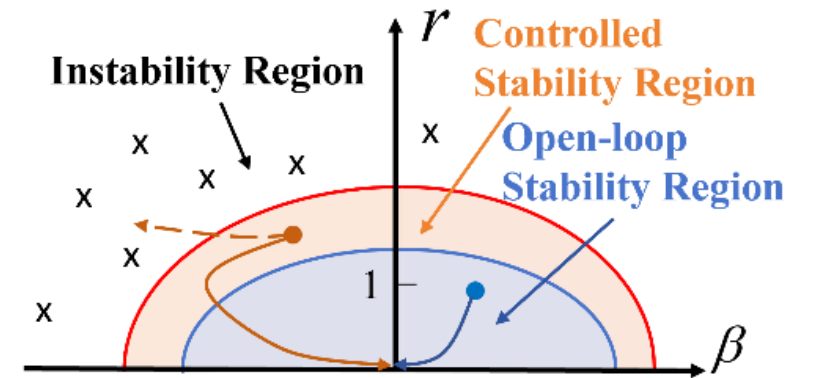
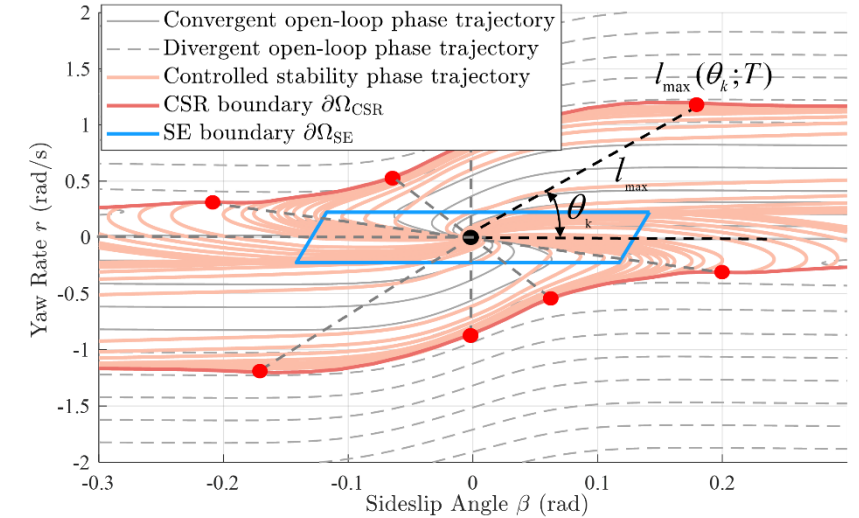
Quantification of Controlled Stability Region

Controlled Trajectory Calculation



Offline
Traversal

Forward-Simulated Radial Bisection

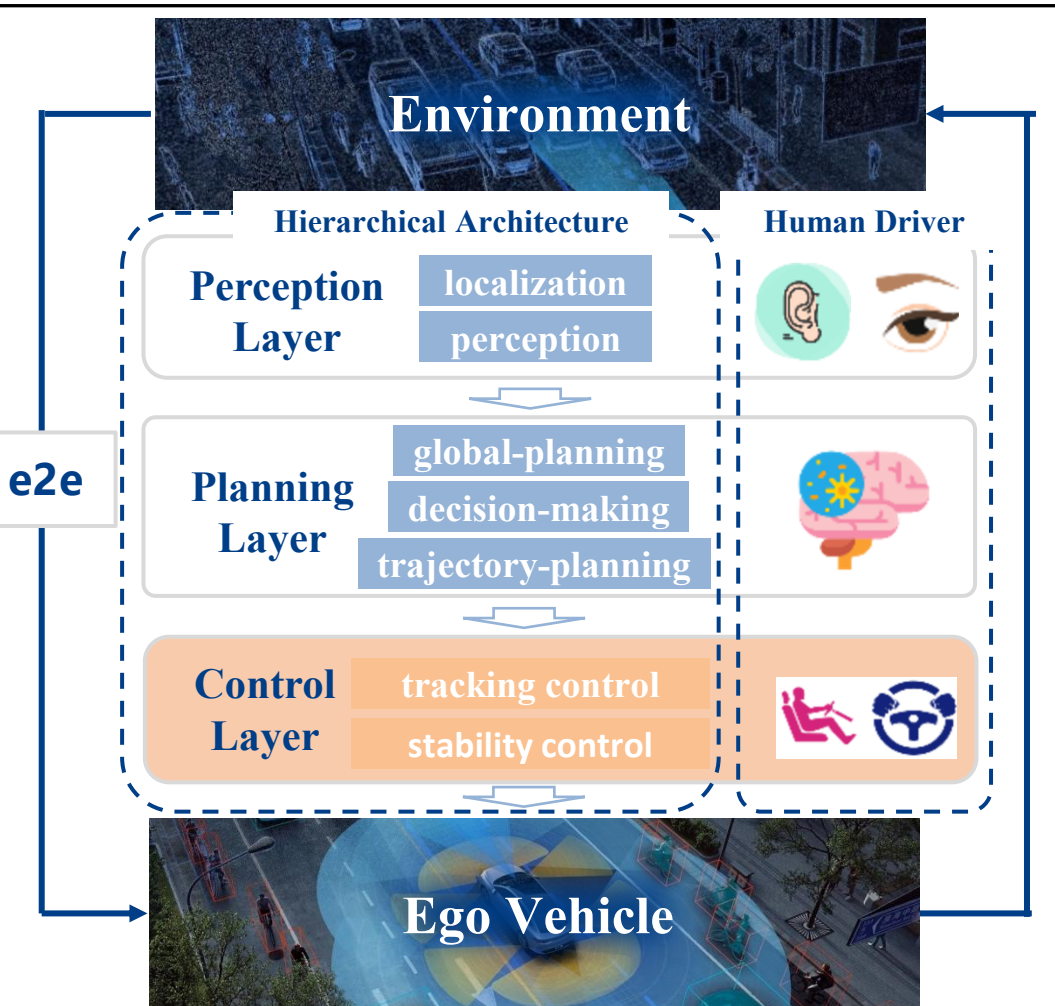


Related publications:

1. **Chengye Wang**, Yu Zhang, Xuepeng Hu, Shaoyang Shi, Jingqi He, Yechen Qin. "Coordinated Tracking and Stability Control of Automated Vehicles at the Limits: A Controlled Stability Region Approach", **2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)**, Naples, Italy. (accepted)
2. **Chengye Wang**, Shaoyang Shi, Yu Zhang, and Yechen Qin. "Controlled Stability Region-Guided Lateral-Longitudinal Trajectory Tracking for Autonomous Vehicles." **IEEE Transactions on Industrial Electronics**. (Under review, IF=7.4, JCR Q1, Top)

Research Background 2: Coordinated Tracking and Stability Control

■ Challenge of Motion Control

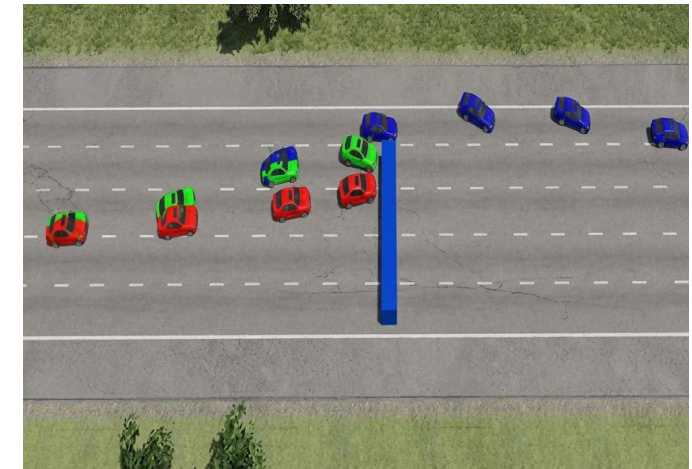


Stability Guarantee

Tracking Accuracy Enhancement



Safety Risk: Lack of strict stability bounds leads to loss of control.

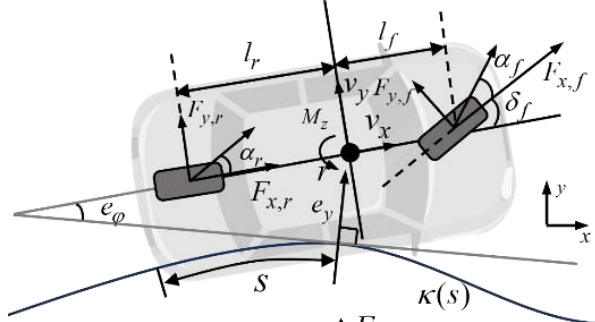


Maneuverability Loss: Over-conservative stability criteria limit maneuverability in extreme conditions.

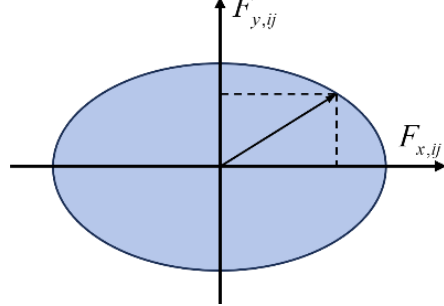
The Tracking-Stability Coordination is the Core Challenge in Motion Control

Research Experience 2: Coordinated Tracking and Stability Control

Longitudinal-Lateral Trajectory Tracking Controller



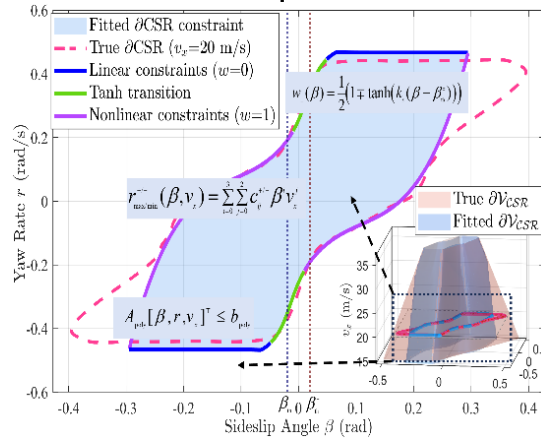
NMPC Predictive Model



Combined-slip Model

$$\left(\frac{F_{x,ij}}{\mu F_{z,ij}} \right)^2 + \left(\frac{F_{y,ij}}{\mu F_{z,ij}} \right)^2 \leq 1$$

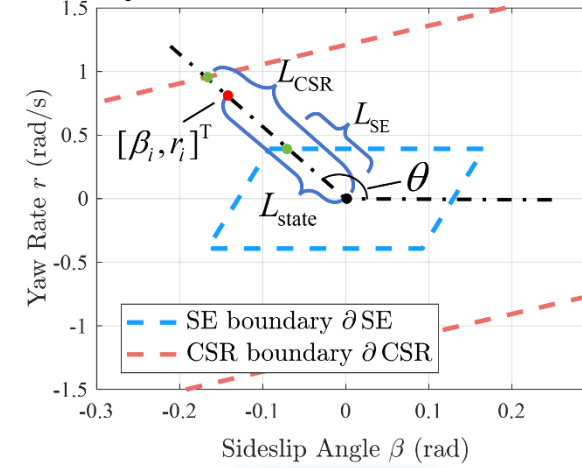
Stability Constraints



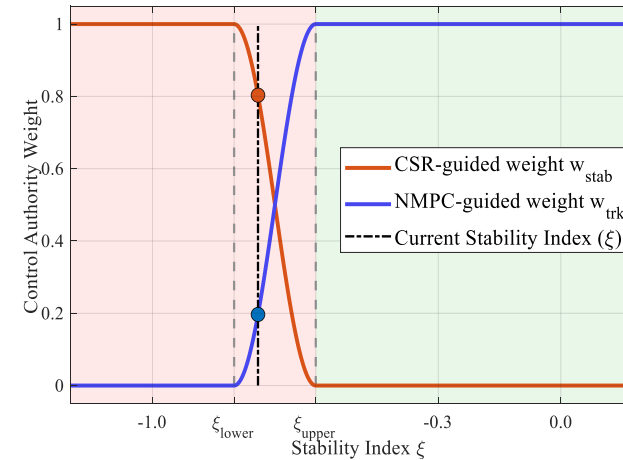
$$\begin{aligned} u^{\text{trk}*} &= \arg \min J^{\text{trk}} \\ \text{s.t. } \dot{x}^{\text{trk}} &= f(x^{\text{trk}}, u^{\text{trk}}), h(x^{\text{trk}}, u^{\text{trk}}) \leq 0 \\ u_{\min}^{\text{trk}} &\leq u^{\text{trk}} \leq u_{\max}^{\text{trk}}, x_{\min}^{\text{trk}} \leq x^{\text{trk}} \leq x_{\max}^{\text{trk}} \\ x_0^{\text{trk}} &= \phi(t_c; x_{\text{meas}}, u_{\text{prev}}^{\text{trk}*}) \end{aligned}$$

CSR-Guided Stability Fallback and Switching Strategy

Instability risk calculation:



Switching Strategy:



$$u^{\text{fianl}}(t) = (1 - w_{\text{stab}}) u^{\text{trk}*}(t) + w_{\text{stab}} u^{\text{stab}*}(t)$$



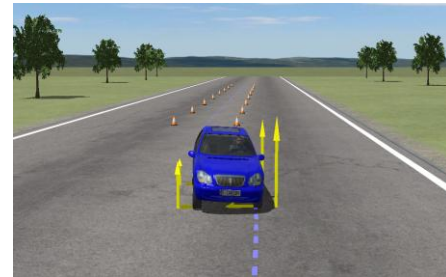
Steer



Drive



Brake



■ Hardware-in-the-Loop (HiL) Platform

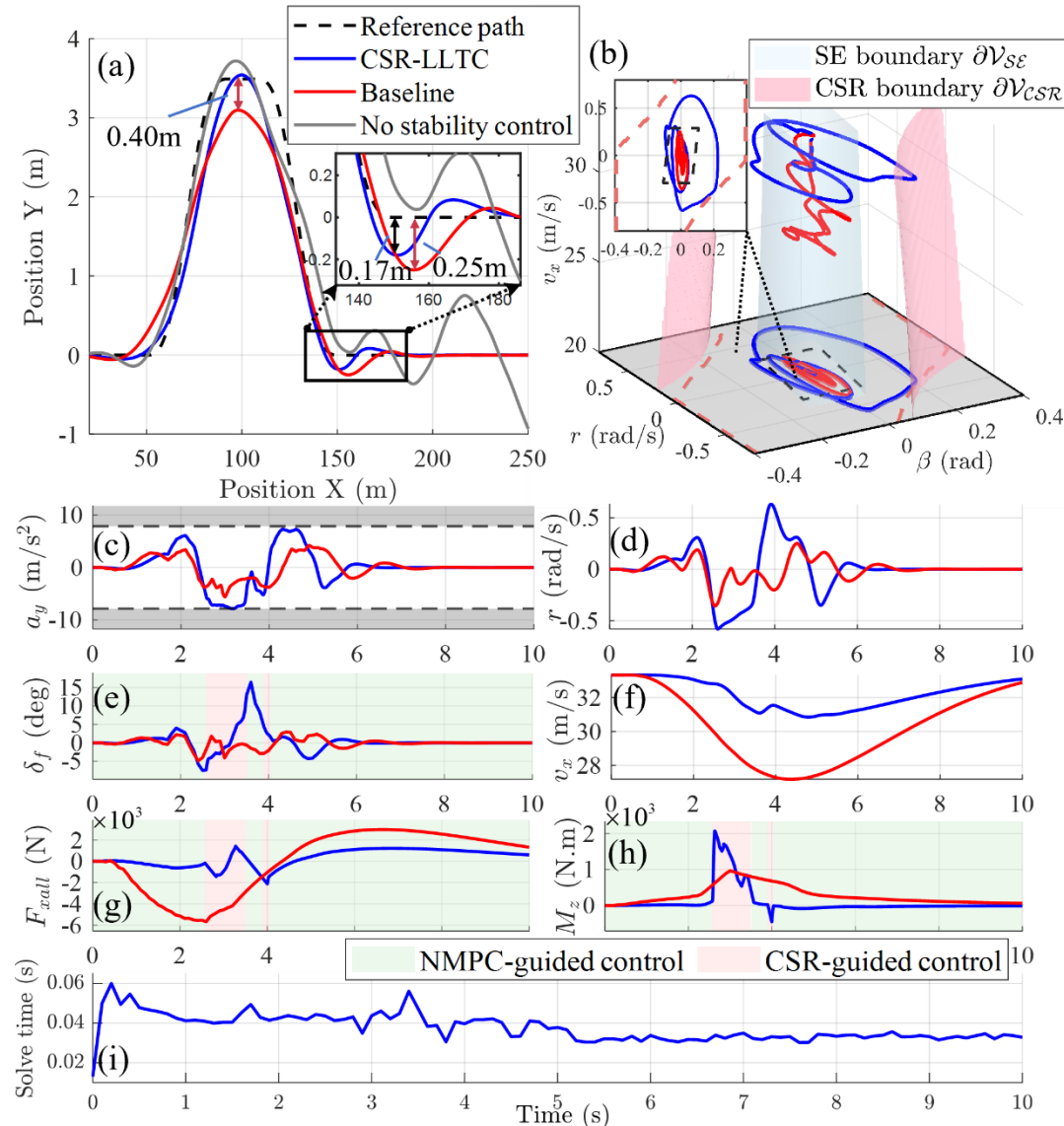


PARAMETERS OF THE TEST VEHICLE

Parameter	Value	Parameter	Value
m	2045 kg	r_w	0.345 m
I_z	4095 kg · m ²	l_f, l_r	1.265, 1.895 m
I_w	2 kg · m ²	B_f, B_r	1.605 m

Research Experience 2: Coordinated Tracking and Stability Control

Double Lane Change (DLC) under $\mu=0.8$, $v_{x,ref}=120\text{km/h}$



Blue car: proposed

Red car: Baseline

Green car: no stability control

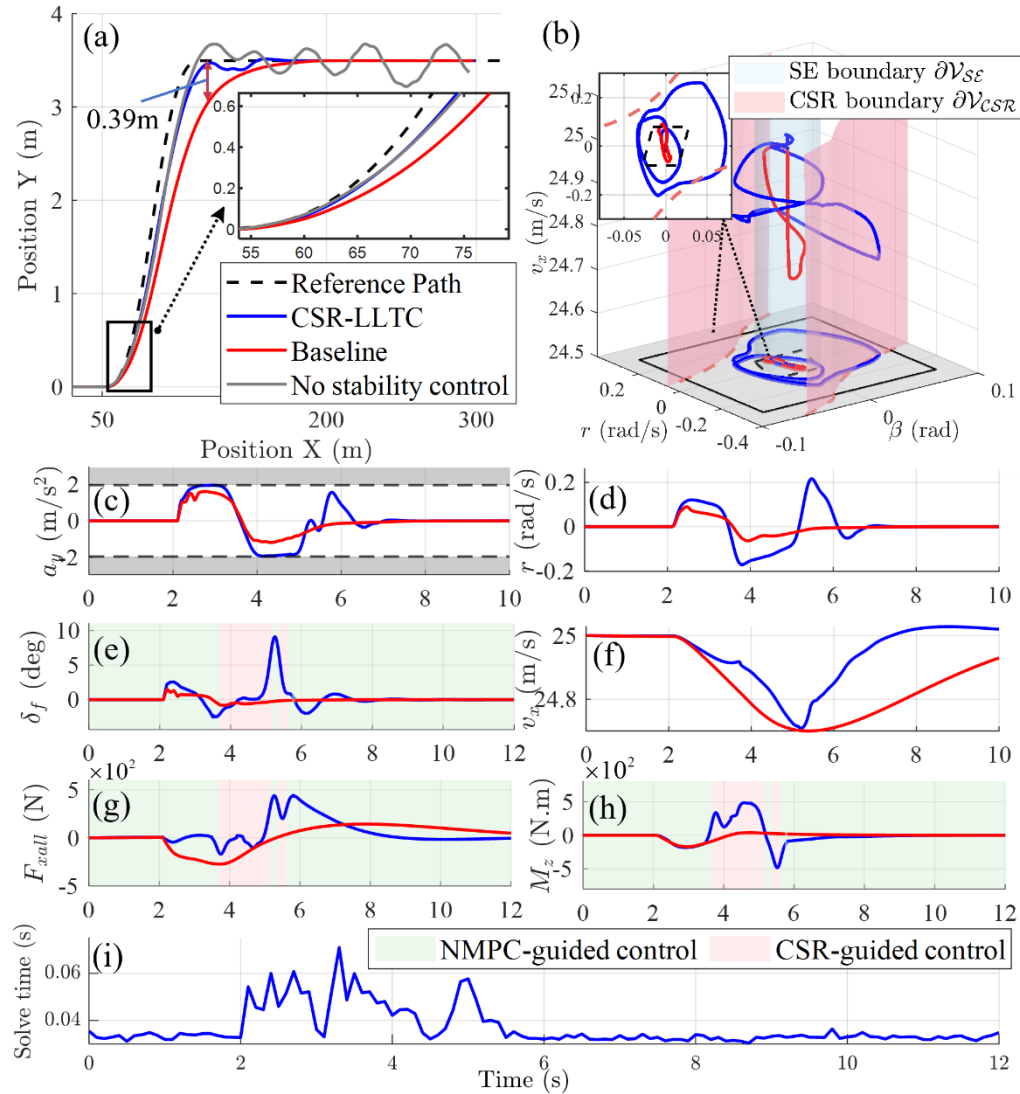
$$e_{y,pk} \downarrow 36.6\%$$

$$v_{x,avg} \uparrow 7.5\%$$



Research Experience 2: Coordinated Tracking and Stability Control

■ Lane Change (LC) under $\mu=0.2$, $v_{x,ref}=90\text{km/h}$



Blue car: proposed

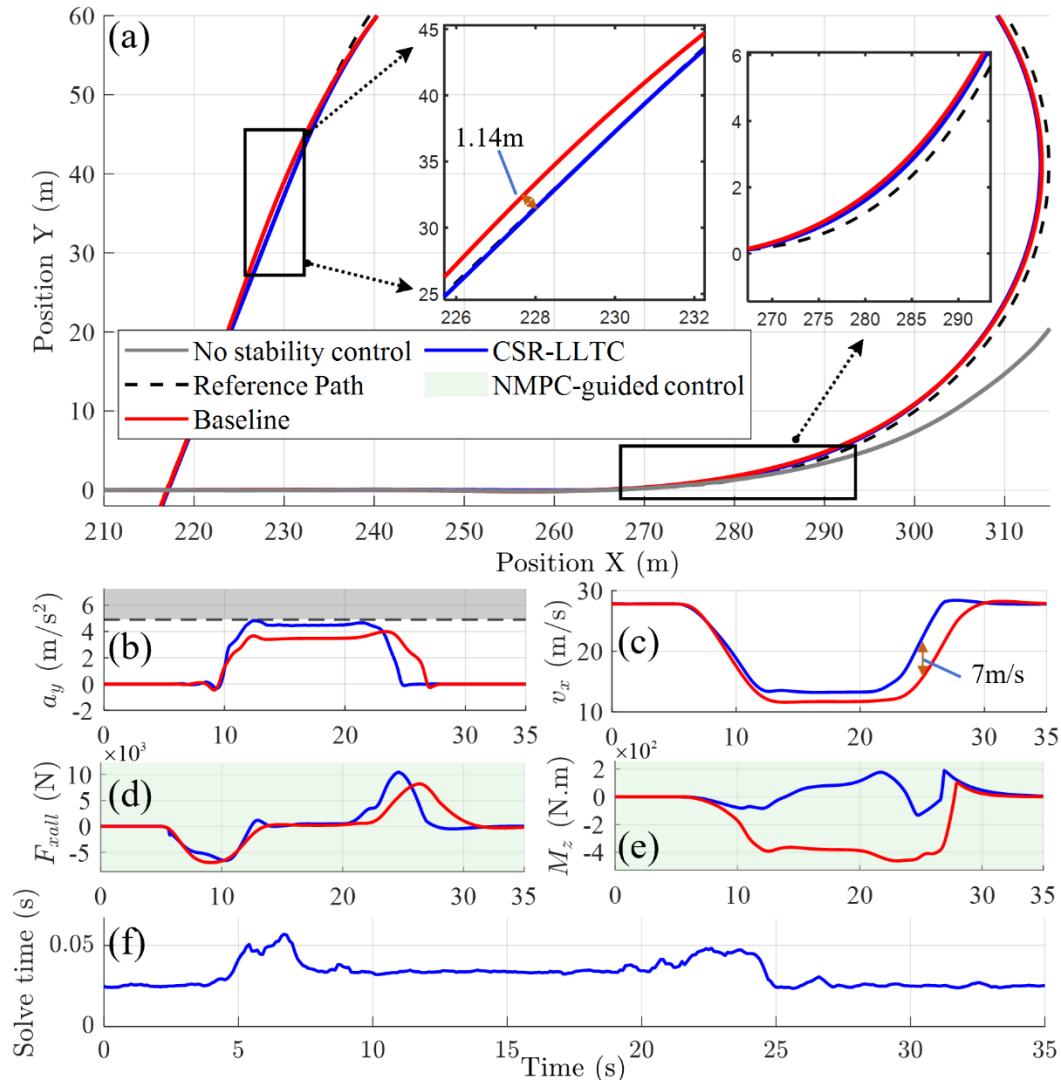
Red car: Baseline

Green car: no stability control



$$e_{y,pk} \downarrow 31.3\%$$

■ Race-Track U-Turn Test under $\mu=0.5$, $v_{x,ref}=100\text{km/h}$



$v_{x,avg} \uparrow 9.3\%$

Related publications:

1. **Chengye Wang**, Yu Zhang, Xuepeng Hu, Shaoyang Shi, Jingqi He, Yechen Qin. "Coordinated Tracking and Stability Control of Automated Vehicles at the Limits: A Controlled Stability Region Approach", **2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)**, Naples, Italy. (accepted)
2. **Chengye Wang**, Shaoyang Shi, Yu Zhang, and Yechen Qin. "Controlled Stability Region-Guided Lateral-Longitudinal Trajectory Tracking for Autonomous Vehicles." **IEEE Transactions on Industrial Electronics**.(Under review, IF=7.4, JCR Q1, Top)

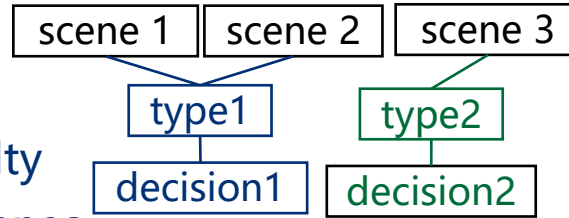
Research Background 3: Model-data-driven trajectory planning

■ Challenge of Trajectory Planning

Decision-Making Method Research

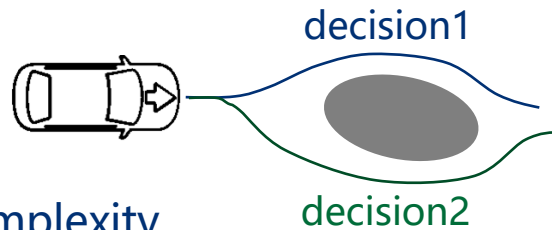
➤ Rule-based

- Easy for deployment
- Exhaustive search difficulty and hard for dynamic scenes



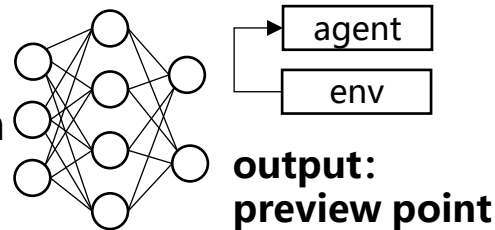
➤ Planning-based

- linked with planning
- High computational complexity



➤ Learning-based

- Changing scenarios adaption
- Poor interpretability



A Framework Integrates Multiple Methods

Trajectory Planning Method Research

➤ Sampling-based

- Strong exploration ability
- Slow obstacle avoidance response

➤ Graph search-based

- Theoretical completeness
- Hard to balance the real-time and quality

➤ Curve interpolation-based

- High efficiency
- Lack of dynamical feasibility

➤ Optimization-based

- Advantage in high-quality optimal trajectories.
- Contradiction between dynamic accuracy and computational efficiency

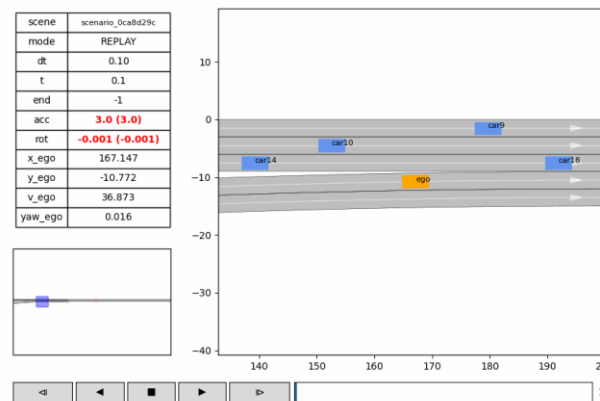
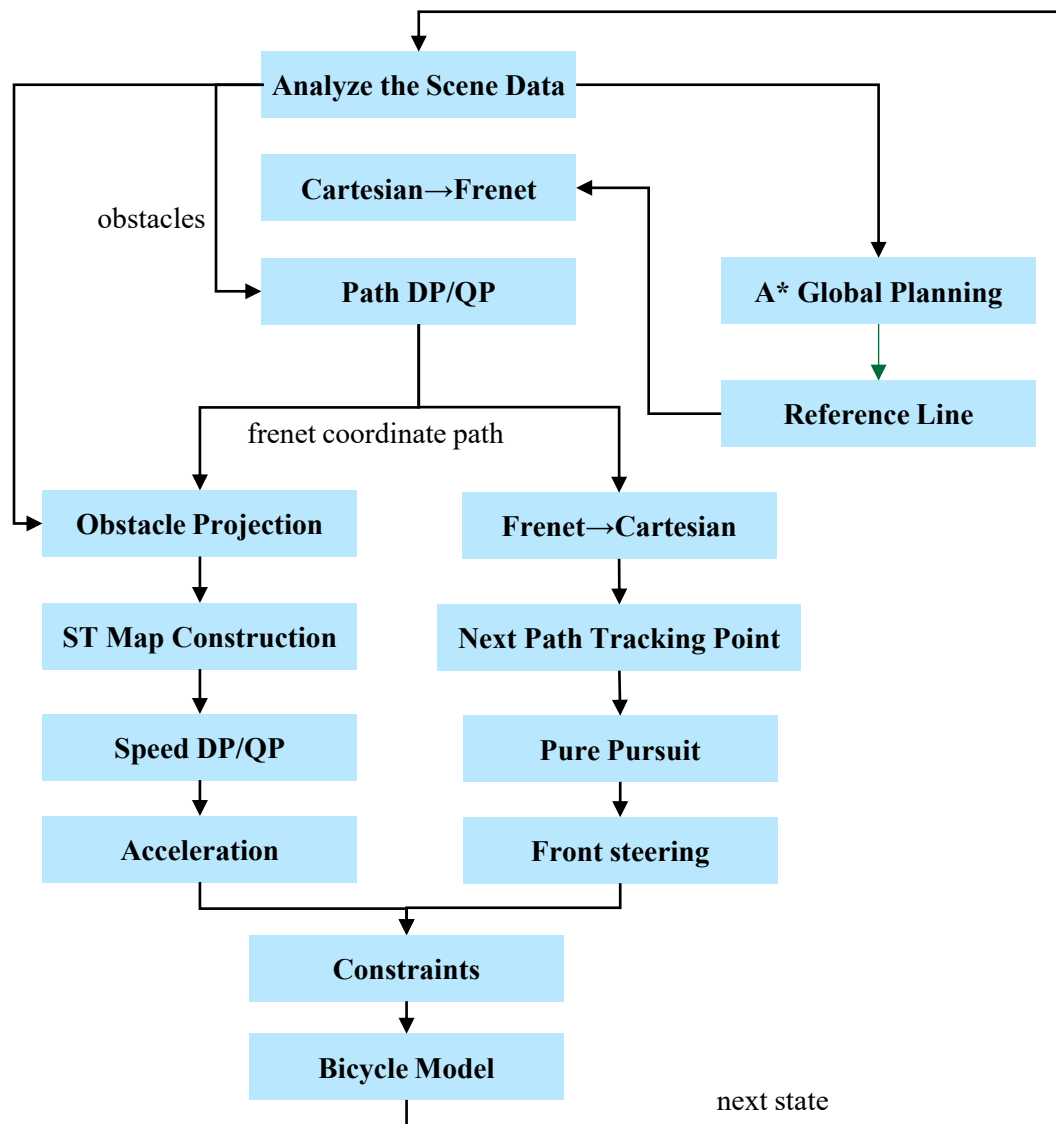
➤ Learning-based

- High efficiency
- Poor interpretability

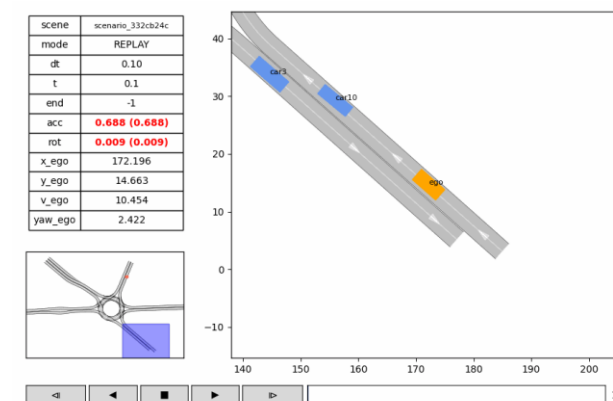
Data-enhanced model optimization method

The Need to Integrate Model-based Approach with the Data-driven Methods

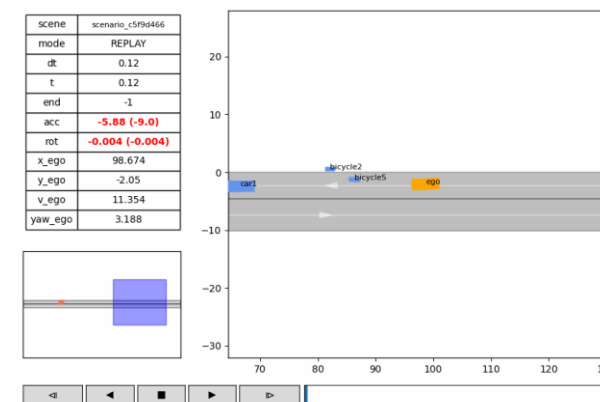
4th in Onsite Autonomous Driving Algorithm Challenge 2025



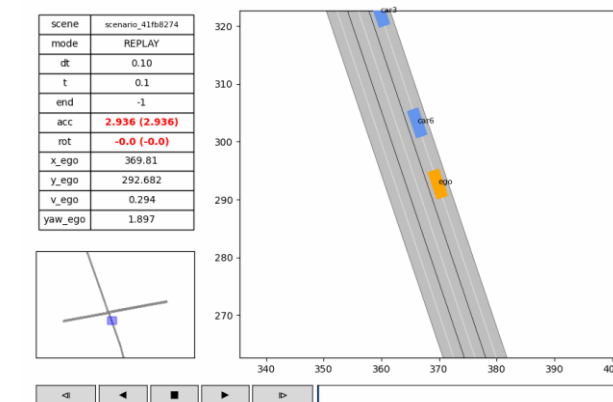
Highway Driving and Highway Merging Scenarios



Urban Roundabout Scenario



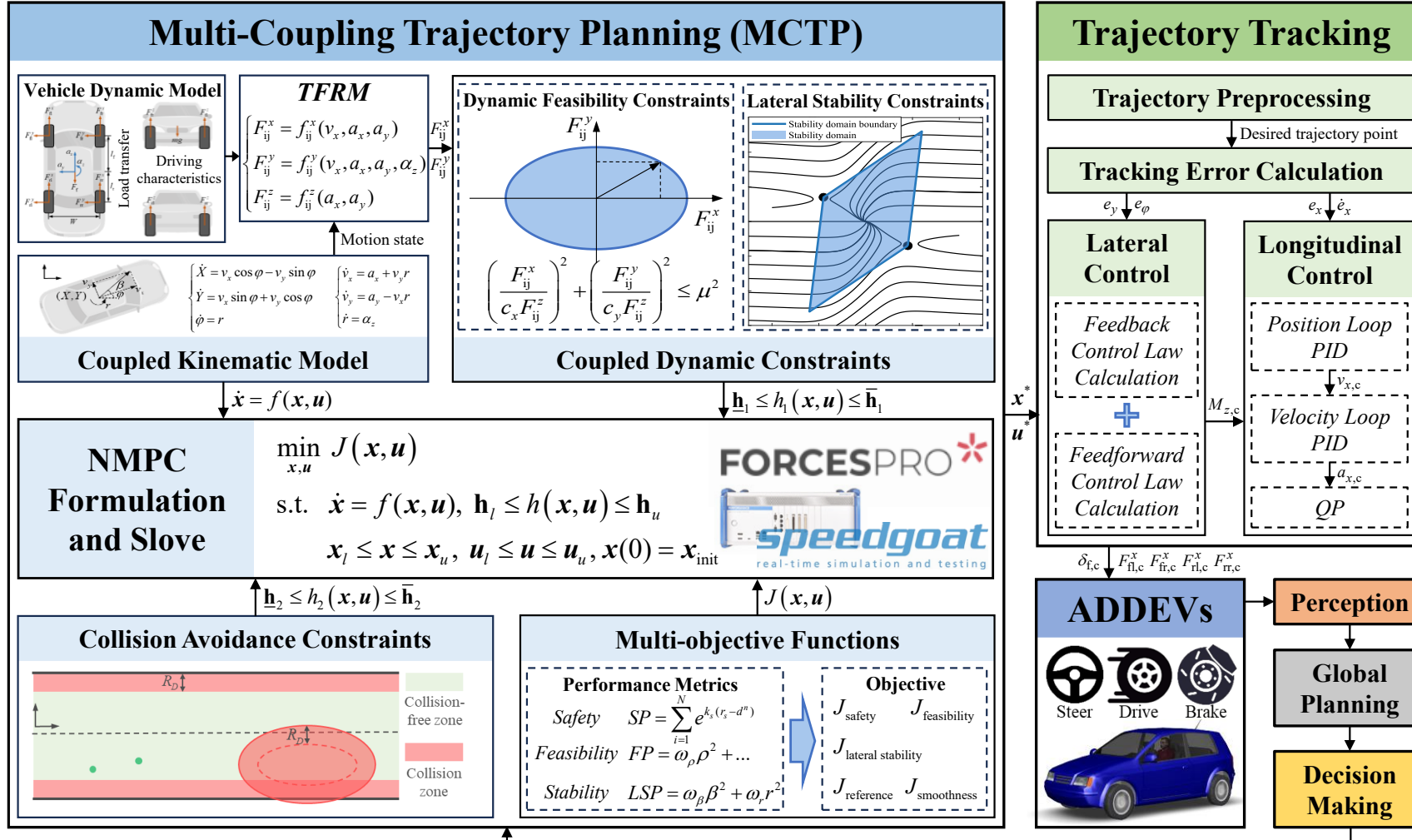
Urban Mixed Traffic Scenario with Bicycles and Pedestrians



Urban Intersection Scenario

Research Experience 3: Model-data-driven trajectory planning

■ Optimization-based Trajectory Planning Framework

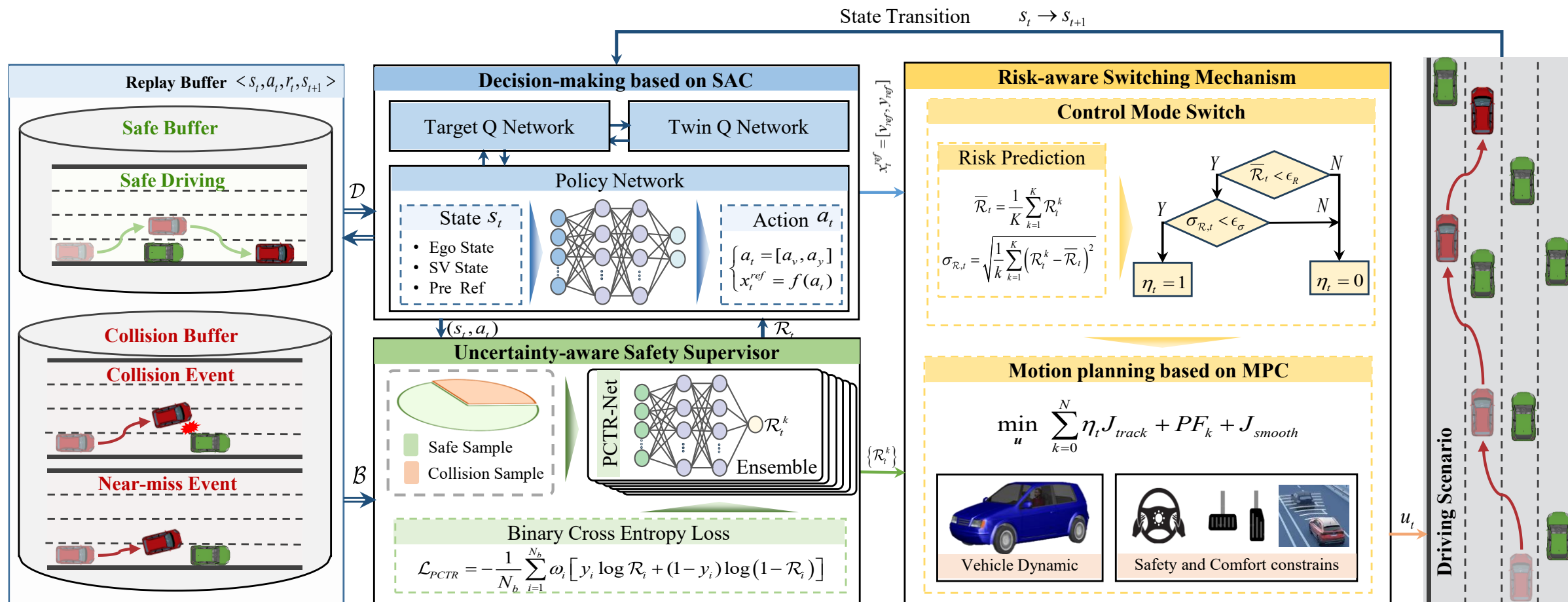


MCTP: A Multi-Coupled Dynamics Trajectory Planning Scheme for Autonomous Driving in Extreme Conditions

Related publications:

1. Xuepeng Hu, Yu Zhang, **Chengye Wang**, Zhenfeng Wang, and Yechen Qin. "MCTP: A Multi-Coupled Dynamics Trajectory Planning Scheme for Autonomous Driving in Extreme Conditions." **IEEE Transactions on Automation Science and Engineering**. (accepted, Q2, Top)

Model-data-driven Framework for Autonomous Driving



Related publications:

1. Shaoyang Shi, Kai Yang, **Chengye Wang**, Dongwei Li, Yu Zhang and Yechen Qin. "Self-Evolving Reinforcement Learning Framework for Autonomous Driving via Uncertainty-Aware Collision-Risk Supervision" **IEEE Transactions on Automation Science and Engineering** .(Under review, IF=6.4, JCR Q2, Top)
2. Shaoyang Shi, Yu Zhang, Kai Yang, **Chengye Wang**, Zhenfeng Wang, and Yechen Qin. "A Risk-Aware Decision Making and Motion Planning Framework for Interactive Autonomous Driving." **2026 IEEE 29th International Conference on Intelligent Transportation Systems (ITSC)**, Naples, Italy. IEEE. (Accepted)

■ Model-data-driven Framework for Autonomous Driving



Self-Evolving Reinforcement Learning Framework
for Autonomous Driving
via Uncertainty-Aware Collision-Risk Supervision



Wheel-End Actuation

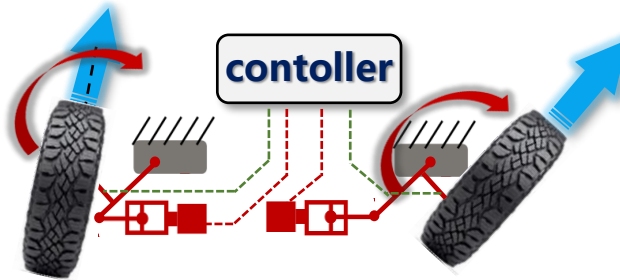


Wheel-end Driving, Braking, and Steering



Power-by-Wire
Ultimate Efficiency

Independent Wheel Control

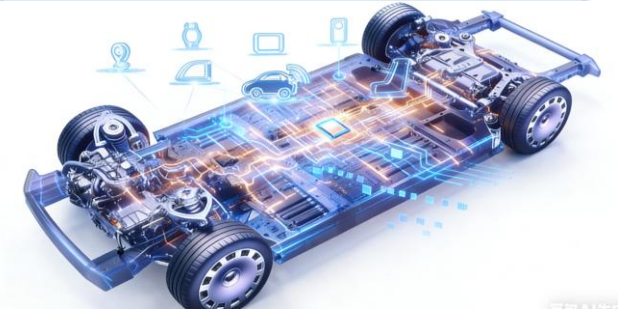


Tire-Road Adhesion → Extremum

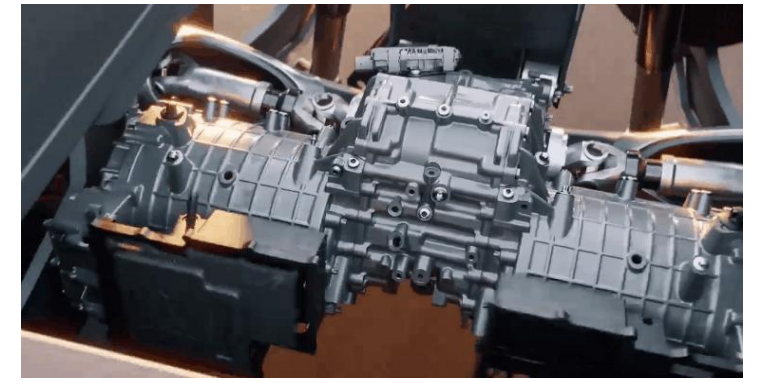


High Control DOFs
Extended Dynamic Boundaries

Software-Defined

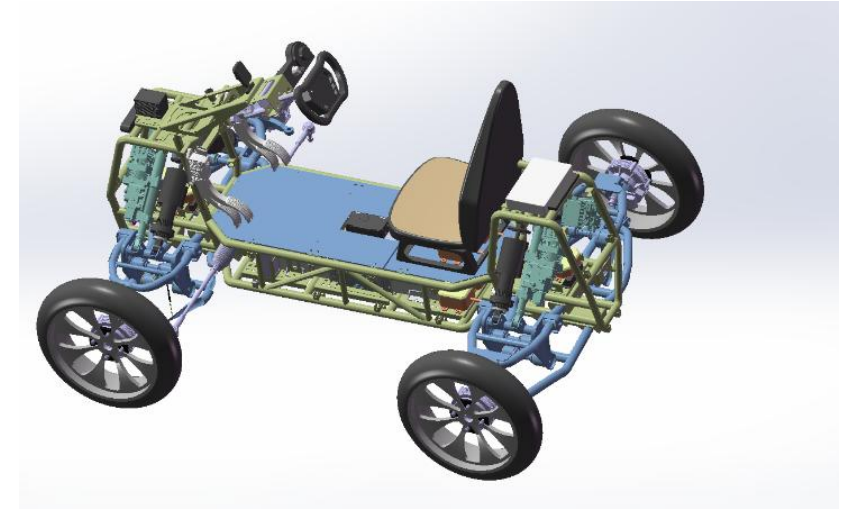


AD-Chassis Domain Integration,
Customizable Functions



The Optimal Platform for
Autonomous Driving

DDEV Architecture Unleashes the Potential of Vehicle Motion Control



In-Wheel Motor



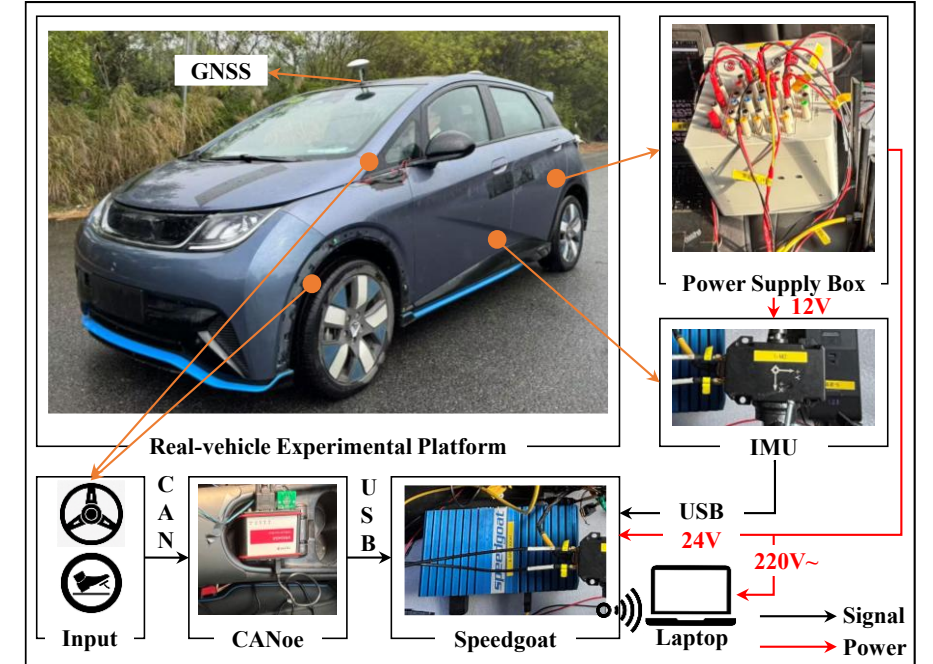
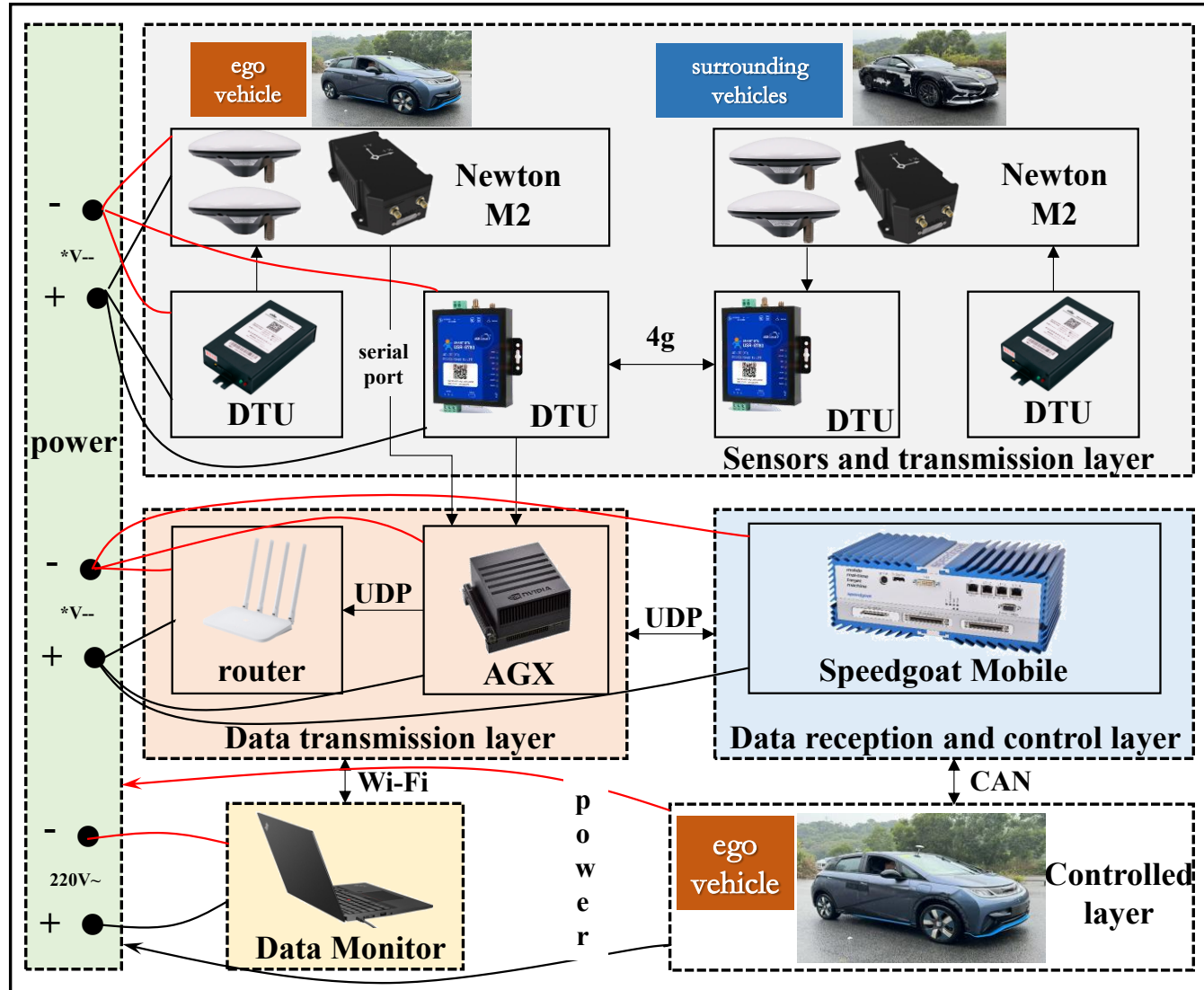
Controller-ECU



IMU&GPS

4WD Electric Vehicle

Research Experience 4: BYD Real-vehicle test



1. Positioning modules and IMU analysis based on Ubuntu/ROS
2. Experimental data communication and hardware debugging

- Learning-based MPC framework under extreme conditions (low adhesion, high speed)
- Model-data-driven decision-making and trajectory planning for autonomous vehicles.



Talk, SAE 2025 Intelligent and Connected Vehicles Symposium, Chongqing, China.



Presentation on behalf our lab to Prof. Amir Khajepour, University of Waterloo, Canada.



Presentation coming soon. See you in ITSC 2026.



Thanks!

Chengye Wang

Supervisor: Prof. Yechen Qin

Email: leowang1100@gmail.com

Homepage: <https://leowang1820.github.io/>

Beijing, China